ADVANCED SAMPLE SURVEY DESIGN

Professional Training Manual

Statistical Domain | Advanced Level | Approximately 6 hours

Audience: Government officials and public-sector professionals

Proposed Table of Contents

1. Course Overview

Course Purpose

Advanced Sample Survey Design develops practical understanding of stratified, multi-stage and unequal-probability sample designs, weighting, estimation and uncertainty. The course treats survey sampling as an integrated process: define the inference target, assess the frame, select units using a defensible design, collect data consistently, estimate population quantities and communicate uncertainty.

Why This Subject Matters to Government Officials

Public programmes often need reliable information about populations that cannot be observed completely or continuously. A defensible sample can reduce cost and burden while supporting useful estimates, provided that coverage, selection, response, measurement and estimation are addressed.

Element	Key concern
Objective	What exactly must be estimated or described?
Target population	Who should the estimates describe?
Frame	Who can actually be selected?
Selection design	How does each unit enter the sample?
Fieldwork	Are selected units contacted and measured consistently?
Estimation	How are observations translated to population statistics?
Uncertainty	How is sampling variability represented?

Audience, Prerequisites, Duration and Level

Audience: Government officials and public-sector professionals. **Prerequisites:** basic probability, descriptive statistics and familiarity with survey terminology. **Level:** Advanced. **Duration:** approximately 6 hours.

2. Learning Objectives

1. Explain advanced principles of probability sample design and estimation.
2. Identify complexities arising from multi-stage, clustered and unequal-probability designs.
3. Apply principles of stratification, allocation and selection probabilities.
4. Compare designs using precision, cost, operational burden and estimation requirements.
5. Analyze coverage, non-response, measurement and sampling-error risks.
6. Design an integrated sampling plan for a public-sector requirement.
7. Evaluate weighting, calibration and variance-estimation concepts.
8. Interpret design effects, standard errors and confidence intervals.
9. Select quality controls and documentation requirements.
10. Document assumptions, limitations and methodological decisions.

3. Course Structure

Module	Title	Topics	Time	Expected outcome
1	Survey Design Architecture	objective and estimands; target population; frame; units; lifecycle	45 min	Explain the architecture of an advanced survey.
2	Stratification and Allocation	strata; proportional/other allocation; domains; subgroup precision	60 min	Plan stratification and allocation.
3	Multi-Stage and Unequal-Probability Sampling	PSUs; SSUs; clusters; unequal probabilities; PPS concepts	75 min	Plan a multi-stage probability design.

Module	Title	Topics	Time	Expected outcome
4	Sample Size and Precision	precision; design effects; domains; non-response; effective sample size	60 min	Connect sample planning to precision.
5	Weighting and Estimation	base weights; non-response adjustment; calibration; estimators	60 min	Develop an estimation framework.
6	Variance Estimation and Quality Assurance	variance; standard errors; confidence intervals; design-aware methods	60 min	Evaluate uncertainty and quality.
—	Exercises and Assessment	Exercises, case study and assessment	60 min	Demonstrate integrated understanding.

4. Detailed Modules

Module 1: Survey Design Architecture

Learning Objectives

• Explain advanced principles of probability sample design and estimation.

• Identify complexities arising from multi-stage, clustered and unequal-probability designs.

• Apply principles of stratification, allocation and selection probabilities.

• Compare designs using precision, cost, operational burden and estimation requirements.

• Analyze coverage, non-response, measurement and sampling-error risks.

Key Concepts

Sampling is a method of selecting a subset of a defined population to obtain information about that population. The quality of inference depends on the relationship among objective, population, frame, selection design and estimation.

Detailed Explanation

Define the statistical objective and estimand: State the population quantity or descriptive result the survey must produce.

Define target population and units: Specify who is in scope and what entity is observed.

Assess frame coverage and quality: Check omissions, duplicates, outdated records and out-of-scope units.

Choose design and allocation: Select a probability design appropriate to inference, cost and operational constraints.

Select units according to documented probabilities: Use a reproducible random/controlled procedure and retain selection records.

Monitor response and fieldwork: Track contact outcomes, non-response and deviations from protocol.

Construct weights and estimate: Use selection probabilities and documented adjustments to support population estimates.

Assess uncertainty and quality: Evaluate sampling variability and relevant non-sampling risks.

Report definitions and limitations: State populations, denominators, methods, uncertainty and important limitations.

Important Terminology

inclusion probability, base weight, stratum, primary sampling unit, cluster, design effect, non-response adjustment, calibration, variance estimation, effective sample size.

Step-by-Step Process

1. Define objective and key estimates

2. Define population and unit

3. Assess frame

4. Select design

5. Plan sample size/allocation

6. Select sample

7. Conduct fieldwork

8. Process and validate

9. Estimate and assess uncertainty

10. Document and report

Illustrative Public-Sector Example

A department needs estimates for several regions. It first defines the target population, then assesses whether available registers cover all relevant users before choosing a sample design.

Common Mistakes

• Choosing units because they are convenient.

• Assuming the frame is complete without checking it.

• Treating sample size as proof of representativeness.

• Ignoring unequal probabilities or clustering in estimation.

• Failing to document non-response and deviations.

Good Practices

• Maintain a written design specification.

• Keep frame and instrument versions.

• Pilot selection and field procedures.

• Monitor disposition and differential response.

• Retain selection, weight and estimation documentation.

• Report limitations clearly.

Short Case Study

A team starts with a questionnaire instead of an objective. Reviewers require an explicit estimand and population definition before sampling.

Module Summary: Begin with the inference target.

Module 2: Stratification and Allocation

Learning Objectives

• Explain advanced principles of probability sample design and estimation.

• Identify complexities arising from multi-stage, clustered and unequal-probability designs.

• Apply principles of stratification, allocation and selection probabilities.

• Compare designs using precision, cost, operational burden and estimation requirements.

• Analyze coverage, non-response, measurement and sampling-error risks.

Key Concepts

The sampling frame is the operational representation used for selection. Probability sampling assigns known or calculable inclusion probabilities. Common designs include simple random, systematic, stratified and cluster or multi-stage sampling.

Detailed Explanation

Define the statistical objective and estimand: State the population quantity or descriptive result the survey must produce.

Define target population and units: Specify who is in scope and what entity is observed.

Assess frame coverage and quality: Check omissions, duplicates, outdated records and out-of-scope units.

Choose design and allocation: Select a probability design appropriate to inference, cost and operational constraints.

Select units according to documented probabilities: Use a reproducible random/controlled procedure and retain selection records.

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Step-by-Step Process

1. Define objective and key estimates
2. Define population and unit
3. Assess frame
4. Select design
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9. Estimate and assess uncertainty
10. Document and report

Illustrative Public-Sector Example

A service register contains geographic information but misses some digital users. The team assesses coverage and considers a combined frame or a restricted inference population.

Common Mistakes

• Choosing units because they are convenient.

• Assuming the frame is complete without checking it.

• Treating sample size as proof of representativeness.

• Ignoring unequal probabilities or clustering in estimation.

• Failing to document non-response and deviations.

Good Practices

• Maintain a written design specification.

• Keep frame and instrument versions.

• Pilot selection and field procedures.

• Monitor disposition and differential response.

• Retain selection, weight and estimation documentation.

• Report limitations clearly.

Short Case Study

A frame contains duplicates and missing geographic areas. The team cleans and documents the frame before selection.

Module Summary: The frame determines who can be selected.

Module 3: Multi-Stage and Unequal-Probability Sampling

Learning Objectives

• Explain advanced principles of probability sample design and estimation.

• Identify complexities arising from multi-stage, clustered and unequal-probability designs.

• Apply principles of stratification, allocation and selection probabilities.

• Compare designs using precision, cost, operational burden and estimation requirements.

• Analyze coverage, non-response, measurement and sampling-error risks.

Key Concepts

The sampling frame is the operational representation used for selection. Probability sampling assigns known or calculable inclusion probabilities. Common designs include simple random, systematic, stratified and cluster or multi-stage sampling.

Detailed Explanation

Define the statistical objective and estimand: State the population quantity or descriptive result the survey must produce.

Define target population and units: Specify who is in scope and what entity is observed.

Assess frame coverage and quality: Check omissions, duplicates, outdated records and out-of-scope units.

Choose design and allocation: Select a probability design appropriate to inference, cost and operational constraints.

Select units according to documented probabilities: Use a reproducible random/controlled procedure and retain selection records.

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Construct weights and estimate: Use selection probabilities and documented adjustments to support population estimates.

Assess uncertainty and quality: Evaluate sampling variability and relevant non-sampling risks.

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Step-by-Step Process

1. Define objective and key estimates
2. Define population and unit
3. Assess frame
4. Select design
5. Plan sample size/allocation
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8. Process and validate
9. Estimate and assess uncertainty
10. Document and report

Illustrative Public-Sector Example

A statewide survey needs regional estimates. Stratification can support regional representation; clustering can reduce field travel but may affect precision.

Common Mistakes

• Choosing units because they are convenient.

• Assuming the frame is complete without checking it.

• Treating sample size as proof of representativeness.

• Ignoring unequal probabilities or clustering in estimation.

• Failing to document non-response and deviations.

Good Practices

• Maintain a written design specification.

• Keep frame and instrument versions.

• Pilot selection and field procedures.

• Monitor disposition and differential response.

• Retain selection, weight and estimation documentation.

• Report limitations clearly.

Short Case Study

A design is selected to balance regional coverage and field cost. The team records the selection probabilities needed for estimation.

Module Summary: Design and selection probabilities connect sampling to estimation.

Module 4: Sample Size and Precision

Learning Objectives

• Explain advanced principles of probability sample design and estimation.

• Identify complexities arising from multi-stage, clustered and unequal-probability designs.

• Apply principles of stratification, allocation and selection probabilities.

• Compare designs using precision, cost, operational burden and estimation requirements.

• Analyze coverage, non-response, measurement and sampling-error risks.

Key Concepts

Sample size is a planning quantity linked to desired precision, key estimates, subgroup requirements, expected response, design effects and available resources. It is not a substitute for a defensible selection design.

Detailed Explanation

Define the statistical objective and estimand: State the population quantity or descriptive result the survey must produce.

Define target population and units: Specify who is in scope and what entity is observed.

Assess frame coverage and quality: Check omissions, duplicates, outdated records and out-of-scope units.

Choose design and allocation: Select a probability design appropriate to inference, cost and operational constraints.

Select units according to documented probabilities: Use a reproducible random/controlled procedure and retain selection records.

Monitor response and fieldwork: Track contact outcomes, non-response and deviations from protocol.

Construct weights and estimate: Use selection probabilities and documented adjustments to support population estimates.

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Step-by-Step Process

1. Define objective and key estimates
2. Define population and unit
3. Assess frame
4. Select design
5. Plan sample size/allocation
6. Select sample
7. Conduct fieldwork
8. Process and validate
9. Estimate and assess uncertainty
10. Document and report

Illustrative Public-Sector Example

Two proposed samples have different subgroup allocations. The team compares expected precision, subgroup requirements, cost and response before choosing the plan.

Common Mistakes

• Choosing units because they are convenient.

• Assuming the frame is complete without checking it.

• Treating sample size as proof of representativeness.

• Ignoring unequal probabilities or clustering in estimation.

• Failing to document non-response and deviations.

Good Practices

• Maintain a written design specification.

• Keep frame and instrument versions.

• Pilot selection and field procedures.

• Monitor disposition and differential response.

• Retain selection, weight and estimation documentation.

• Report limitations clearly.

Short Case Study

A subgroup has a larger planned allocation because reliable subgroup estimates are required; the final estimates must account for the design.

Module Summary: Sample size supports precision but cannot compensate for poor coverage.

Module 5: Weighting and Estimation

Learning Objectives

• Explain advanced principles of probability sample design and estimation.

• Identify complexities arising from multi-stage, clustered and unequal-probability designs.

• Apply principles of stratification, allocation and selection probabilities.

• Compare designs using precision, cost, operational burden and estimation requirements.

• Analyze coverage, non-response, measurement and sampling-error risks.

Key Concepts

Weights connect sampled units to population estimation. The base weight is commonly related to the inverse of inclusion probability. Non-response adjustment and calibration may be appropriate under documented methodology.

Detailed Explanation

Define the statistical objective and estimand: State the population quantity or descriptive result the survey must produce.

Define target population and units: Specify who is in scope and what entity is observed.

Assess frame coverage and quality: Check omissions, duplicates, outdated records and out-of-scope units.

Choose design and allocation: Select a probability design appropriate to inference, cost and operational constraints.

Select units according to documented probabilities: Use a reproducible random/controlled procedure and retain selection records.

Monitor response and fieldwork: Track contact outcomes, non-response and deviations from protocol.

Construct weights and estimate: Use selection probabilities and documented adjustments to support population estimates.

Assess uncertainty and quality: Evaluate sampling variability and relevant non-sampling risks.

Report definitions and limitations: State populations, denominators, methods, uncertainty and important limitations.

Important Terminology

inclusion probability, base weight, stratum, primary sampling unit, cluster, design effect, non-response adjustment, calibration, variance estimation, effective sample size.

Step-by-Step Process

1. Define objective and key estimates

2. Define population and unit

3. Assess frame

4. Select design

5. Plan sample size/allocation

6. Select sample

7. Conduct fieldwork

8. Process and validate

9. Estimate and assess uncertainty

10. Document and report

Illustrative Public-Sector Example

Units have unequal selection probabilities. The estimation plan records inclusion probabilities and uses appropriate weights; additional non-response or calibration adjustments are considered only with documented methodology.

Common Mistakes

• Choosing units because they are convenient.

• Assuming the frame is complete without checking it.

• Treating sample size as proof of representativeness.

• Ignoring unequal probabilities or clustering in estimation.

• Failing to document non-response and deviations.

Good Practices

• Maintain a written design specification.

• Keep frame and instrument versions.

• Pilot selection and field procedures.

• Monitor disposition and differential response.

• Retain selection, weight and estimation documentation.

• Report limitations clearly.

Short Case Study

Response is lower in one subgroup. The team investigates differential non-response and does not assume weighting has eliminated all bias.

Module Summary: Weights represent the sample in population estimation and must be documented.

Module 6: Variance Estimation and Quality Assurance

Learning Objectives

• Explain advanced principles of probability sample design and estimation.

• Identify complexities arising from multi-stage, clustered and unequal-probability designs.

• Apply principles of stratification, allocation and selection probabilities.

• Compare designs using precision, cost, operational burden and estimation requirements.

• Analyze coverage, non-response, measurement and sampling-error risks.

Key Concepts

Survey uncertainty depends on the actual design. Stratification, clustering, unequal probabilities and weighting can affect variance. Design-aware variance estimation is therefore important for complex samples.

Detailed Explanation

Define the statistical objective and estimand: State the population quantity or descriptive result the survey must produce.

Define target population and units: Specify who is in scope and what entity is observed.

Assess frame coverage and quality: Check omissions, duplicates, outdated records and out-of-scope units.

Choose design and allocation: Select a probability design appropriate to inference, cost and operational constraints.

Select units according to documented probabilities: Use a reproducible random/controlled procedure and retain selection records.

Monitor response and fieldwork: Track contact outcomes, non-response and deviations from protocol.

Construct weights and estimate: Use selection probabilities and documented adjustments to support population estimates.

Assess uncertainty and quality: Evaluate sampling variability and relevant non-sampling risks.

Report definitions and limitations: State populations, denominators, methods, uncertainty and important limitations.

Important Terminology

inclusion probability, base weight, stratum, primary sampling unit, cluster, design effect, non-response adjustment, calibration, variance estimation, effective sample size.

Step-by-Step Process

1. Define objective and key estimates
2. Define population and unit
3. Assess frame
4. Select design
5. Plan sample size/allocation
6. Select sample
7. Conduct fieldwork
8. Process and validate
9. Estimate and assess uncertainty
10. Document and report

Illustrative Public-Sector Example

A complex sample uses multiple stages and clusters. Reporting includes design-consistent uncertainty rather than applying a simple-random formula without justification.

Common Mistakes

• Choosing units because they are convenient.

• Assuming the frame is complete without checking it.

• Treating sample size as proof of representativeness.

• Ignoring unequal probabilities or clustering in estimation.

• Failing to document non-response and deviations.

Good Practices

• Maintain a written design specification.

• Keep frame and instrument versions.

• Pilot selection and field procedures.

• Monitor disposition and differential response.

• Retain selection, weight and estimation documentation.

• Report limitations clearly.

Short Case Study

A report initially ignores clustering. The methodological review requires design-aware variance estimation and transparent documentation.

Module Summary: Uncertainty must reflect the actual sample design.

5. Practical Exercises

Exercise 1: Define the Sampling Problem

Choose an illustrative public-sector programme. Define the target population, unit, frame, reference period and three key estimates.

Expected output: One-page sampling specification.

Evaluation: methodological consistency, explicit assumptions, correct terminology, defensible reasoning and clear documentation.

Exercise 2: Compare Designs

Compare simple random, stratified and cluster sampling for a geographically dispersed population. Consider representation, cost, precision and complexity.

Expected output: Design comparison and recommendation.

Evaluation: methodological consistency, explicit assumptions, correct terminology, defensible reasoning and clear documentation.

Exercise 3: Build a Selection Plan

Using a hypothetical frame, specify the selection procedure, sample allocation and documentation required.

Expected output: Selection-plan note.

Evaluation: methodological consistency, explicit assumptions, correct terminology, defensible reasoning and clear documentation.

Exercise 4: Weighting and Variance

Explain base weights, non-response adjustment, calibration and design-aware variance estimation for a hypothetical complex sample.

Expected output: Methodological note.

Evaluation: methodological consistency, explicit assumptions, correct terminology, defensible reasoning and clear documentation.

6. Integrated Public-Sector Case Study

Background

Illustrative scenario: A department needs estimates of service access and completion across several regions. Services are delivered through offices and digital channels. Administrative records exist but coverage differs by channel.

Problem

A proposal suggests interviewing whoever is available at selected offices while also reporting an overall estimate for all users.

Available Information

• Service transaction register

• Geographic information

• Digital transaction list

• Historical response information

• Limited information on abandoned applications

Tasks for the Learner

• Define the inference population and unit.

• Assess frame coverage.

• Compare two sampling designs.

• Specify how regional estimates can be supported.

• Identify non-response risks.

• Describe estimation and uncertainty requirements.

Recommended Approach

First define the population and estimands. If digital and office users are both in scope, the frame must cover both or the inference must be restricted. If regional estimates are required, stratification can be considered. If travel cost is important, multi-stage cluster sampling may be considered. The final weights and variance method must match the design.

Worked Solution

Element	Illustrative solution
Objective	Estimate defined service-access/completion measures for a defined population and period.
Population	Eligible service users or application attempts under explicit rules.

Element	Illustrative solution
Frame	Integrated or appropriate operational sources after coverage assessment.
Design	Stratified or multi-stage probability design, depending on frame, geography and cost.
Non-response	Monitor differential response and investigate causes.
Estimation	Use design-consistent weights and variance estimation.
Reporting	State denominators, definitions, uncertainty and limitations.

7. Assessment

10 Multiple-Choice Questions

1. What determines the intended inference target?
Answer: Statistical objective and target population.

2. Which design divides the population into strata?
Answer: Stratified sampling.

3. If inclusion probability is π, the basic inverse-probability weight is:
Answer: $1/\pi$.

4. A poor frame creates what major risk?
Answer: Coverage error.

5. A larger sample does not automatically eliminate:
Answer: Bias.

6. Why use cluster sampling?
Answer: To potentially reduce field cost.

7. What does design effect compare?
Answer: Variance under a complex design with a reference design.

8. Why stratify?
Answer: To support representation and potentially improve precision.

9. What should accompany a survey estimate?
Answer: Definition, methodology and relevant uncertainty/limitations.

10. Can weighting fix every survey error?
Answer: No.

5 True/False Questions

1. A probability sample uses known or calculable selection probabilities. True.

2. A sampling frame is always identical to the target population. False.

3. Cluster sampling can affect variance relative to simple random sampling. True.

4. Overall response rate alone proves estimates are unbiased. False.

5. Design information should be retained for estimation and uncertainty analysis. True.

5 Short-Answer Questions

1. Define a sampling frame.
Answer: An operational representation from which a sample is selected.

2. What is an inclusion probability?
Answer: The probability that a population unit is included under the design.

3. Why use stratification?
Answer: To support representation and potentially improve precision.

4. What is non-response bias?
Answer: Potential distortion when non-respondents differ systematically from respondents.

5. Why use design-aware variance estimation?
Answer: Because the actual design can change sampling variance relative to simple random sampling.

2 Scenario-Based Questions

Scenario 1: A survey of only office visitors is used to represent office and digital users. **Answer:** Coverage and selection problems arise because digital users are excluded.

Scenario 2: A complex clustered sample is analyzed as simple random sampling. **Answer:** Standard errors and confidence intervals may be incorrect because the actual design is ignored.

8. Final Assignment

Objective

Prepare a defensible sampling-design note for a government programme or public service.

Instructions

1. State information need and estimands.

2. Define population and unit.

3. Assess frame and coverage.

4. Choose and justify design.

5. Specify sample-size/allocation assumptions.

6. Explain selection probabilities or procedure.

7. Describe non-response and quality controls.

8. Specify estimation and uncertainty approach.

9. State limitations.

10. Provide implementation plan.

Suggested time: 90–120 minutes. **Deliverable:** 5–8 page methodological note plus sample-selection outline.

Assessment Rubric

Criterion	Marks
Objective/population/frame	15
Sampling design	25
Allocation/sample size	15
Implementation/quality	15
Estimation/uncertainty	15
Limitations/documentation	10
Clarity	5
Total	100

9. Glossary

Allocation: Distribution of sample across strata or groups.

Base weight: Weight related to inverse selection probability.

Bias: Systematic deviation from the intended target.

Calibration: Adjustment of weights to align with suitable control totals under a documented method.

Census: Collection intended to cover all in-scope units.

Cluster: Group of units sampled together at one or more stages.

Coverage error: Error caused by mismatch between population and operational coverage.

Design effect: Ratio comparing variance under a complex design with a reference design.

Frame: Operational representation used for sample selection.

Inclusion probability: Probability a unit is included in the sample.

Measurement error: Difference between recorded and intended value.

Non-response: Failure to obtain required information.

PSU: Primary sampling unit in a multi-stage design.

Population: Complete set of units relevant to the objective.

Probability sampling: Sampling with known or calculable selection probabilities.

Sample: Subset selected for observation.

Sampling error: Variation from observing a sample rather than the full population.

Stratification: Division into strata before selection.

Target population: Population about which inference is intended.

Unit of observation: Entity for which information is collected.

Variance: Measure of statistical variability/uncertainty.

Weight: Factor used to represent units in population estimation.

10. Quick Reference

Core Workflow

Objective → Population → Unit → Frame → Design → Allocation/Sample Size → Selection → Fieldwork → Processing → Weights → Estimation → Variance → Reporting

Decision Rules

• Define the inference population before choosing the sample.

• Assess frame coverage before selection.

• Use probability sampling when design-based population inference is required.

• Match weights and variance methods to the actual design.

• Investigate differential non-response.

• Document every material design deviation.

Formulae

Inclusion probability: π■. **Basic inverse-probability weight:** w■=1/π■. **Introductory proportion planning:** $n \approx z^2 p(1-p)/e^2$. These formulas require appropriate assumptions; complex designs may need adjustments and specialized variance methods.

11. References

Public methodological references verified for inclusion; they do not imply official affiliation with this course.

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Capturing What Matters: Essential Guidelines for Designing Household Surveys
World Bank, Living Standards Measurement Study, 2021
https://www.worldbank.org/en/programs/lsms/publication/capturing-what-matters

Survey Methods and Practices
Statistics Canada, 2010

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12. Content Verification Notes

General Educational Guidance

The modules, examples, exercises, assessments and decision rules are original educational material based on generally accepted survey-sampling methodology.

Requires Subject-Matter Review Before Operational Use

• Specific sample-size calculations and precision targets.

• Complex multi-stage selection procedures.

• Weighting and calibration specifications.

• Variance-estimation implementation.

• Legal, privacy, confidentiality and security requirements.

• Department-specific classifications, frames and approved procedures.

Assumptions

• Examples are illustrative.

• No particular department, state, survey frame or legal framework was supplied.

• Course duration is approximately six hours.

• Operational implementation requires project-specific review and approvals.

Information That May Vary

Frames, classifications, reference periods, sampling standards, field procedures, legal requirements, data-governance rules and reporting practices may vary by department, state, law, programme and survey purpose.